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BOOK REVIEW

Review of advanced issues in partial least squares structural equation modeling (second edition)

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This article provides a review of the book *Advanced Issues in Partial Least Squares Structural Equation Modeling (2nd Edition)* by Joseph F. Hair Jr., Marko Sarstedt, Christian M. Ringle, and Siegfried P. Gudergan (Hair et al. 2024). The book builds upon the third edition of *A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)* (Hair et al. 2022) by delving into more advanced topics and techniques in partial least squares structural equation modeling (PLS-SEM) regarding model specification, estimation, results assessment, and validation. In addition, the new edition of the book extends the work of its predecessor (Hair et al. 2018) by providing updates, additional details, and new guidelines regarding previously included subject matter (e.g., higher-order constructs, nonlinear effects, importance-performance map analysis), as well as the inclusion of additional topics such as the necessary condition analysis (NCA).

Although the topics in the Advanced Issues book are more advanced than those of the Primer book, the presentations and discussions of concepts and their analyses are done in a straightforward easy to understand manner. This is done by keeping the text approachable and limiting the usage of equations, formulas, and Greek symbols.

The book is divided into six chapters. Chapter 1 provides a brief overview of the origins and evolution of PLS-SEM, as well as an interesting discussion on why PLS-SEM was initially more slowly adopted than its counterpart, covariance-based structural equation modeling (CB-SEM). In addition, the chapter also walks readers through the basics of model specification and model estimation in PLS-SEM. Principles of PLS-SEM are also discussed, such as philosophy of measurement, parameter estimation accuracy, and model estimation implications, as well as how PLS-SEM differs from CB-SEM regarding each of these.

Chapter 2 covers higher-order constructs, first discussing various terminology related to them as well as motivations for their use. Next, different types of higher-order constructs are discussed, as well as several approaches for their specification, estimation, and validation.

Advanced modeling and model assessment make up Chapter 3. Nonlinear relationships are examined first, followed by the modeling of quadratic effects through the use of the latest version of the SmartPLS software (Ringle et al. 2022). The evaluation of nonlinear effects and interpretation of their results are covered next. These are followed by a discussion on the use of confirmatory tetrad analysis (CTA-PLS) to assist with proper measurement model specification and empirically assessing whether a formative or reflective measurement model should be specified. The chapter concludes with information on how to run CTA-PLS in SmartPLS 4, as well as the interpretation of its results.

Chapter 4 covers advanced results illustration. The first main topic discussed is the NCA, along with its purpose and method, as well as the identification of necessary conditions, and special considerations when running this type of analysis. The chapter also shows readers how to run an NCA in SmartPLS 4 and interpret the results. The chapter's second main topic is the importance-performance map analysis (IPMA), which adds a visual interpretation in order to enhance standard PLS-SEM results reporting. Readers are provided with a step-by-step walkthrough of how to conduct an IPMA using SmartPLS 4, while also conducting several requirements checks along the way.

Modeling observed heterogeneity is the focus of Chapter 5. First, the importance and consequences of heterogenous data are discussed. This is followed by an overview of observed and unobserved heterogeneity. A three-step process of testing for measurement model invariance is covered next. Multigroup analysis (MGA) is then discussed with special attention being paid to parametric MGA, bootstrap MGA,

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permutation MGA, comparing more than two groups, and additional recommendations for conducting a multigroup analysis.

Chapter 6, the book's final chapter, tackles the modeling of unobserved heterogeneity. Two methods that researchers can concurrently use to uncover and treat unobserved heterogeneity in PLS path models are presented. These methods are finite mixture PLS (FIMIX-PLS) and prediction-oriented segmentation in PLS (PLS-POS). The chapter then provides a five-step procedure for assessing unobserved heterogeneity using each of the two methods in SmartPLS 4.

Each chapter includes convenient and detailed rules of thumb regarding its respective topics and methods, as well as a case study to demonstrate the application of the chapter's concepts to a real-world problem. In addition, a single case study is used throughout the book, as opposed to a different case for each chapter. This allows for consistency between chapters and increases understandability for readers, since they do not have to learn the basics of a new case for each chapter. Instead, each chapter builds upon a case they already know. The same case is also used in the Primer book, which provides additional cohesiveness between the two pieces. Furthermore, a single software package, SmartPLS 4, is used to illustrate how to conduct the analyses described in the book, which further increases the book's straightforwardness and approachability. The end of each chapter concludes with a detailed and useful summary, review and critical thinking questions, key terms, and a list of suggested readings for those interested in diving deeper into the chapter's respective topics.

In summation, *Advanced Issues in Partial Least Squares Structural Equation Modeling (2nd Edition)* is an extremely useful and valuable resource for researchers, practitioners, faculty, and students alike from a variety of disciplines looking to enhance their knowledge and abilities regarding PLS-SEM.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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